

Ganyu Xu

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Education

U of Waterloo, MSc in Electrical and Computer Engineering, GPA 94.6/100	Sep 2023 – Aug 2025
U of California, Berkeley, BA in Mathematics and Statistics, GPA: 3.464/4.0	Sep 2015 – May 2019

Technical skills

Programming languages: C, Python 3, Rust, Bash, SQL

Cryptography: Post Quantum Cryptography, Public Key Infrastructure (PKI), Transport Layer Security (TLS)

Systems: Linux/Unix, memory safety (AddressSanitizer), secret-dependent branching detection (Valgrind memcheck), fuzzing

Experience

Cryptography Research Architect, Open Quantum Safe – Waterloo, ON, Canada Aug 2025 – Present

- Core maintainer of **liboqs**, an open-source C library for quantum-resistant cryptographic algorithms.
- Adapted reference cryptographic implementations for integration into liboqs: symmetric primitive shimming, memory safety, constant-time validation, fuzzing, and correctness tests (NIST KAT, ACVP vectors, Wycheproof).
- Triaged and remediated security vulnerabilities and C/Assembly implementation errors, including compiler-induced cache-timing side channel, incorrect incremental absorption in SHA-3 vectorization, and under-allocation for AES key schedule.
- Maintained continuous integration pipeline on GitHub Actions, leveraging static analyzers (CodeQL) and coding agents to automate repetitive maintainanc and uphold code quality.
- Led release process and coordinated vulnerability response through GitHub Security Advisories.

Senior data engineer, LeanTaaS Inc. – Santa Clara, CA, United States Jul 2019 – Sep 2023

- Deployed and maintained distributed Apache Airflow cluster on AWS Elastic Container Service.
- Improved data pipeline observability and production outage response time w/ DataDog and PagerDuty
- Mentored junior developers through project design review, pair programming, and technical workshops

Projects

Post-quantum TLS on embedded devices xuganyu96/embedded-pqtls

- Designed and implemented a post-quantum key encapsulation mechanism based on ML-KEM with an alternative CCA transformation, achieving IND-1CCA security and reducing decapsulation time by more than 50%
- Expanded WolfSSL's support for post-quantum PKI and TLS handshake. Benchmarked performance on Pi Pico 2
- Implemented KEMTLS on WolfSSL, including KEM-based certificate generation and authentication protocol

RustCrypto github.com/RustCrypto

- Contributed to Marvin Attack mitigation in RustCrypto/RSA: ported random prime generation to constant-time heap-allocated integers and integrated Marvin test harness for timing variability detection

Apache Airflow github.com/apache/airflow

- Built **AirflowFargateExecutor**, a task executor that launches ephemeral Fargate containers per task for cost-efficient parallelism
- Fixed improper exception handling in python multiprocessing that causes critical process to become zombie instead of exiting upon database disconnect
- Implemented user-specified database check retries in the CLI command `airflow db check`